

Exercises for Quantum Information

Sheet 1 – Linear Algebra

Basic linear algebra

Problem 1. (1) What is the inner product between the real vectors $(0, 1, 0, 1)$ and $(0, 1, 1, 1)$?
(2) What is the inner product between the quantum states $|0101\rangle$ and $|0111\rangle$?

Tensor products

Quantum states